microeconomics
→ producer (firm)
→ max profit.

* Profit = TR - TC
(Q×P) - (Cost of input)
L ↙ ↘ K
labor → wages (w)
capital → interest (r)

∴ profit = (Q×P) - (w×L + r×K)

* Goal of any firm:
- Max profit
- min Cost.

→ production function:
Q = f(L, K)
↓ function
[depends on]

Sh-run prod.	Long Run prod.
L: variable	L: variable
K: fixed	K: variable
$Q = f(L, \bar{K})$	Q = f(L, K)
* Law of diminishing m.p (Return) → marginal product	* Law of Returns to scale.

* in short run prod., Q depends on L only.

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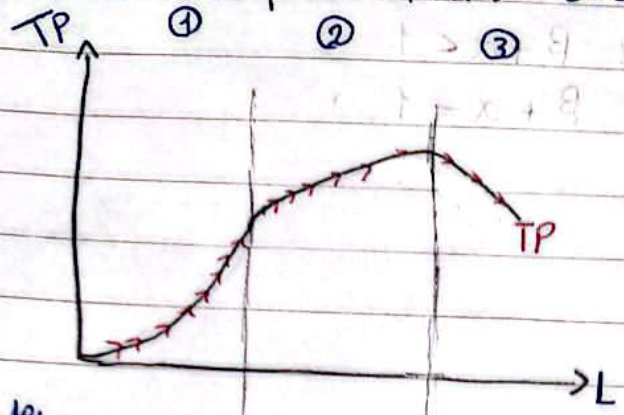
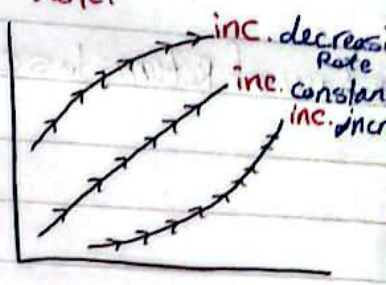
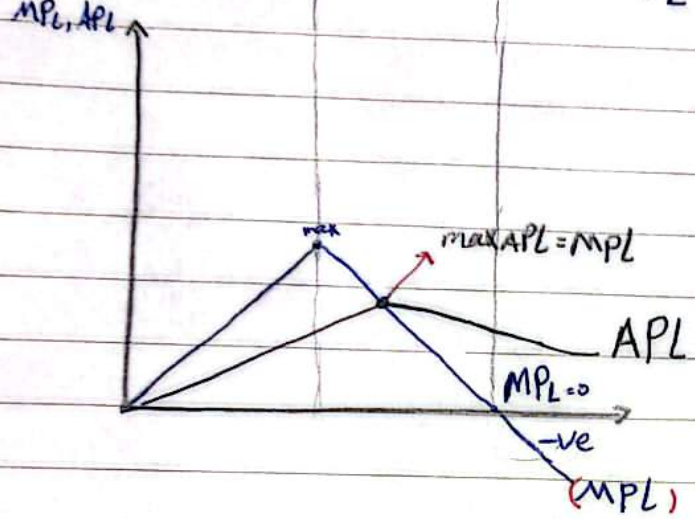
* short run production: 3-stages



Note:





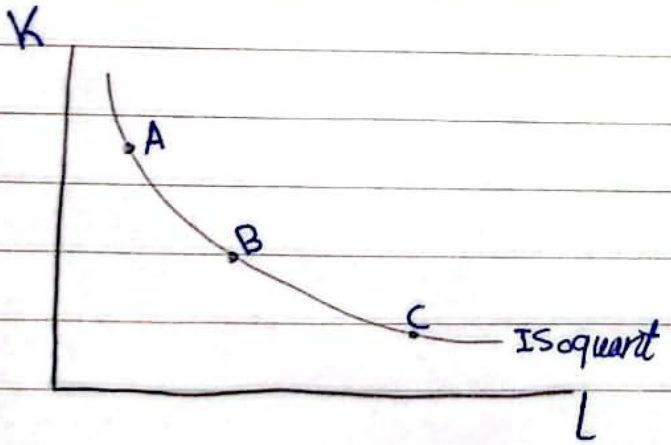
* $MP_L = \frac{\Delta TP}{\Delta L}$

* $AP_L = \frac{TP}{L}$

* Statge 2 : statge of diminishing Retu

* Long Run production:. Iso.quant curve

$Q = f(L, K)$.



* A, B, C ⟹ give the same outpu

* Slope (MRTS) = $\frac{\Delta K}{\Delta L} = \frac{MPL}{MPK}$

marginal rate of technical substitution

⟹ * Cobb-douglas production function:

$Q = A \cdot L^{\beta} \cdot K^{\alpha}$

A ↓ level of technology

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* Law of Returns to scale
  - increasing RTS ($\beta + \alpha > 1$)
  - decreasing RTS ($\beta + \alpha < 1$)
  - Constant RTS ($\beta + \alpha = 1$)

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